

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	1.10.2026	Duration:	60 minutes

Code of Conduct

I E. (1921260160) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

- 1. Understand the Problem Statement**
 - o Explain the problem in your own words.
 - o Identify the objectives of the proposed system.
 - o State the scope of the problem.
- 2. Identify Key Stakeholders and Challenges**
 - o Identify all the stakeholders involved in the system.
 - o Explain the roles and responsibilities of each stakeholder.
 - o Discuss the major challenges and issues faced by the stakeholders.
- 3. Analyze the Current Approach**
 - o Describe the existing system or current process.
 - o Identify its strengths and limitations.
 - o Analyze the problems or gaps that need to be addressed.
- 4. Propose a Suitable Solution Using Software Engineering Principles**
 - o Design a software-based solution for the given problem.
 - o Describe the major functional modules of the proposed system.
 - o Identify the functional and non-functional requirements.

- o Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**
- o Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
 - o Justify why the selected methodology/framework is suitable for the proposed system.
 - o Explain how it supports successful project development and maintenance.
6. **Discuss Expected Outcomes and Limitations**
- o Describe the expected benefits of the proposed solution.
 - o Explain how the solution addresses the identified challenges.
 - o Discuss potential risks, implementation challenges, and limitations.
 - o Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

	(modularity, scalability, maintainability, security, etc.).			
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

AI- Based fake identity & document screening system

Problem Statement

Explain the problem in your own words.

Fake identity documents such as Aadhaar cards, PAN cards, passports, driving licences, and educational certificates can be used for identity theft, financial fraud, unauthorized access, and other illegal activities. Traditional document verification mainly depends on manual checking, which is time-consuming and may miss sophisticated fake documents. Identify the objectives of the proposed system.

The main objectives of the Proposed System

1. To automatically verify identity documents using AI-based techniques.
2. To detect forged, altered, or suspicious documents.
3. To extract important information from documents using OCR.
4. To compare document information with the submitted user details.
5. To perform face matching between the document photo and the user's live/selfie image.
6. To reduce manual verification time and effort.
7. To improve the accuracy and consistency of identity screening.
8. To provide a secure and reliable verification process.
9. To generate alerts for suspicious or high-risk documents.
10. To maintain verification records for authorized users and auditing.

Scope of the Problem

The system can be used in organizations where identity and document verification are required.

Applications

- Banking and financial services
- Insurance companies
- Educational institutions
- Government service portals
- E-commerce and online marketplaces
- Employee onboarding
- Hostel/admission verification
- Telecom customer verification
- Online KYC processes

The system can initially support commonly used identity documents and can later be expanded to support additional document types and verification databases.

2. Identify Key Stakeholders and Challenges

Identify all the stakeholders involved in the system.

The major stakeholders of the Honey Chain system are:

1. Useful for banks, colleges, insurance, government services, companies, KYC, and online platforms.
2. 1. Key Stakeholders & Challenges

3. Users: Submit documents and personal details.
4. Verification Officers: Review results and suspicious cases.
5. Administrators: Manage users and system.
6. Organization: Uses the system for verification.
7. AI System: Performs document and identity analysis.
8. Challenges: Fake documents, manual errors, privacy issues, poor image quality, and high verification workload.

Explain the roles and responsibilities of each stakeholder.

The stakeholders of the AI-Based Fake Identity & Document Screening System include users, verification officers, administrators, organizations, AI systems, security teams, and database administrators. Users upload their identity documents and provide accurate personal details. Verification officers review AI-generated results and investigate suspicious documents. Administrators manage users, system settings, and access permissions. Organizations use the system for identity verification. The AI system extracts document information, detects possible fraud, and generates risk scores. The security team protects sensitive information from unauthorized access, while the database administrator securely stores, manages, and backs up verification data.

3. Analyze the Current Approach

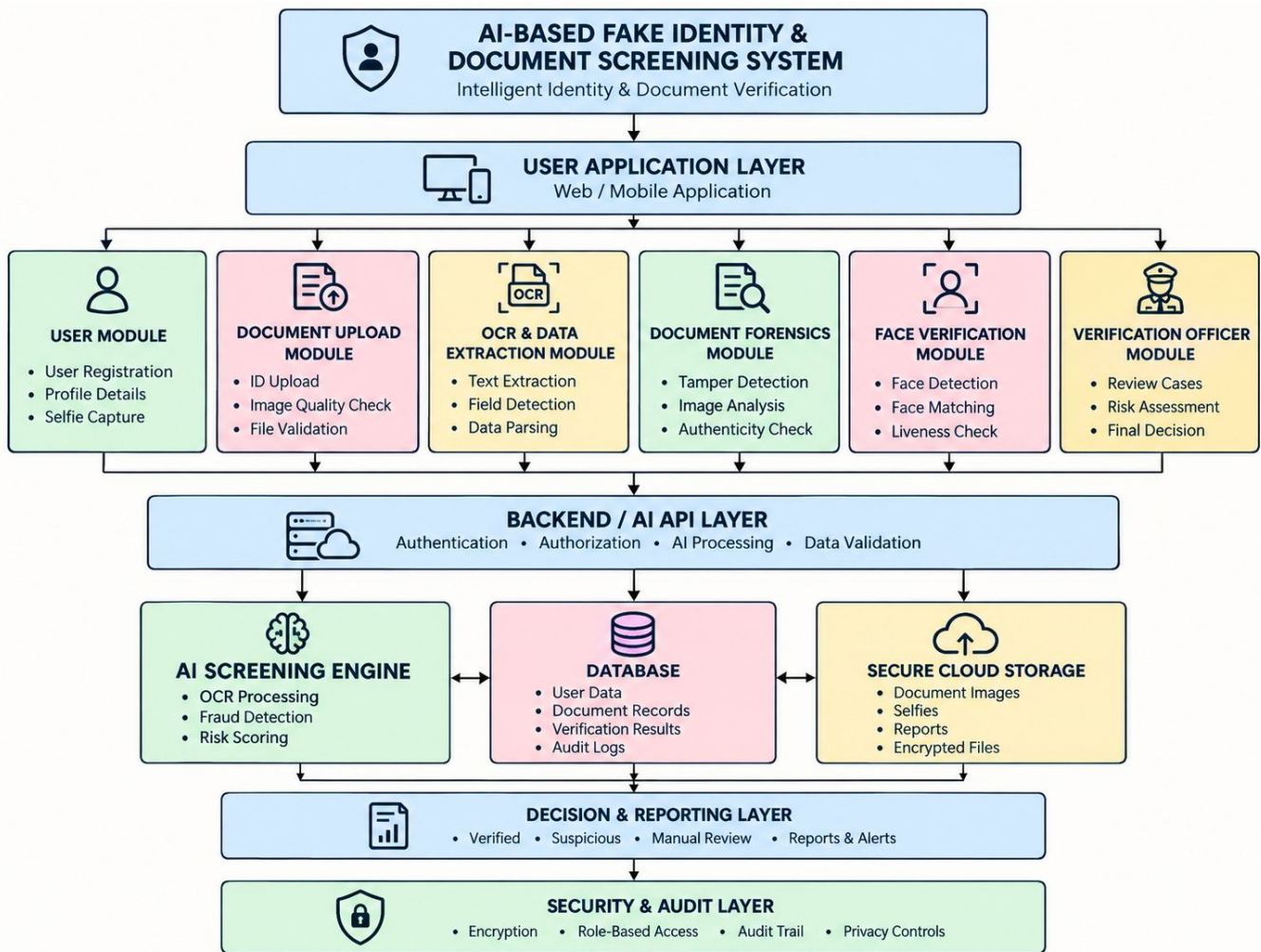
- Existing System: Manual verification of identity documents.
- Process: Staff check documents and compare the given details.
- Strength: Simple and allows human judgment.
- Limitation: Time-consuming and requires more manpower.
- Problem: Human errors may occur during verification.
- Security Gap: Sophisticated fake documents may not be easily detected.
- Scalability Issue: Difficult to process a large number of documents.
- Need: An AI-based system can provide faster, automated, and more reliable screening.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

- Develop a web-based AI system for automatic identity and document screening.
- Users upload their identity documents and personal details.
- OCR extracts information from the uploaded document.
- AI-based document analysis checks for possible alterations or suspicious features.
- Face verification compares the document photo with the user's selfie.
- The system generates a risk score and identifies suspicious submissions.
- Verification officers can review flagged cases through an admin dashboard.
- All verification results are securely stored with access control and audit logs.

Architecture Block



Simple Data Flow – AI-Based Fake Identity & Document Screening System

User

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Register & Enter Details

↓

Upload Identity Document

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OCR – Extract Document Data

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Document Analysis

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Face Matching with Selfie

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AI Risk Assessment

↓

Risk Score Generated

↓

Suspicious?

↓

Yes → Verification Officer Review

No → Verified / Approved

↓

[Store Verification Record](#)

↓

[Admin Dashboard & Report](#)

Describe the major functional modules of the proposed system.

1 Major Functional Modules of the Proposed System

1. User Registration and Authentication Module:

This module allows users and verification officers to securely register and log in. Role-based access control ensures that users can access only the functions relevant to their responsibilities.

2. Document Upload Module:

Users can upload identity documents such as Aadhaar, PAN card, passport, or driving licence. The system checks the file format and image quality before processing.

3. OCR and Data Extraction Module:

OCR extracts important information such as name, date of birth, document number, and address from the uploaded document.

4. Document Analysis Module:

The system analyzes the document for possible tampering, alterations, missing information, and other suspicious features using AI and image-processing techniques.

5. Face Verification Module:

The system compares the photograph on the identity document with the user's selfie to check whether they belong to the same person.

6. AI Fraud Detection Module:

The AI model analyzes document and identity verification results and generates a risk score to identify potentially suspicious submissions.

7. Verification Officer Module:

Verification officers can review flagged documents, check AI results, and make the final verification decision for uncertain or high-risk cases.

8. Database and Record Management Module:

This module securely stores user details, document information, verification results, risk scores, and audit records.

9. Admin and Monitoring Module:

Administrators can manage users, permissions, verification activities, reports, and system performance while maintaining security and privacy controls.

Identify the Functional and Non-Functional Requirements

Functional Requirements:

- User registration and login.
- Role-based access control.
- Identity document upload.
- Document image quality checking.
- OCR-based data extraction.
- Document authenticity and tampering detection.
- Face matching with user selfie.
- AI-based fraud detection.
- Risk score generation.
- Suspicious document flagging.
- Verification officer review.
- Verification result generation.
- Secure storage of user and document records.
- Report generation.
- Administrator management.

Non-Functional Requirements:

- Security: Protect personal and document data from unauthorized access.
- Performance: Process documents and generate results quickly.
- Reliability: Provide consistent and accurate screening results.
- Scalability: Support increasing numbers of users and documents.
- Usability: Provide a simple and user-friendly interface.
- Maintainability: Allow easy updates and maintenance.
- Privacy: Ensure secure handling and appropriate retention of identity data.
- Availability: Keep the system accessible with minimal downtime.

Explain How Software Engineering Principles Are Incorporated

Modularity is achieved by dividing the system into independent modules such as user management, document upload, OCR, document analysis, face verification, AI fraud detection, and reporting.

Scalability is supported through a layered architecture and cloud-based deployment. More users, documents, and verification requests can be handled as the system grows.

Maintainability is achieved through modular source code, well-defined APIs, proper documentation, testing, and separation of frontend, backend, AI model, and database components.

Reliability is supported through data validation, error handling, backup mechanisms, audit logs, and multiple verification methods.

Usability is incorporated through simple dashboards, clear instructions, easy document uploading, and understandable verification results.

Security is implemented through authentication, role-based access control, data encryption, secure APIs, input validation, and controlled access to sensitive identity documents.

5. Justify the Chosen Methodology

Agile Scrum methodology is chosen for the AI-Based Fake Identity & Document Screening System because the project involves multiple modules such as OCR, document analysis, face verification, and AI-based fraud detection. Agile allows the system to be developed module by module through short iterations (sprints). Continuous testing and feedback help improve the AI model and system accuracy. It also makes it easier to handle changing requirements, identify errors early, and deliver a reliable and secure verification system.

Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

6. Discuss Expected Outcomes and Limitations

Describe the expected benefits of the proposed solution.

- **Faster Verification:** Reduces the time required to verify identity documents.
- **Fake Document Detection:** Identifies suspicious or manipulated documents using AI and image analysis.
- **Improved Accuracy:** OCR and face matching reduce manual verification errors.
- **Reduced Workload:** Automates repetitive document screening tasks.
- **Risk Assessment:** Generates risk scores to identify potentially fraudulent applications.
- **Better Security:** Protects sensitive identity information through authentication and access control.
- **Centralized Records:** Stores verification details and results in a secure database.
- **Decision Support:** Helps verification officers make faster and more informed decisions.

Discuss potential risks, implementation challenges, and limitations.

- **False Results:** AI may incorrectly identify genuine documents as fake or miss sophisticated fake documents.
- **Poor Image Quality:** Blurred, damaged, or low-resolution documents can reduce OCR and detection accuracy.
- **Data Privacy:** Identity documents contain sensitive personal information and must be securely stored and processed.
- **AI Bias:** The system may produce inaccurate results if the training data is limited or unbalanced.
- **Implementation Cost:** Developing and maintaining AI models, servers, and security infrastructure can be costly.
- **Integration Challenges:** Connecting the system with existing databases and verification platforms may be difficult.
- **Human Verification:** High-risk or uncertain cases may still require manual verification by authorized officers.
- **Model Maintenance:** AI models need regular updates and retraining to detect new types of document fraud.

Future Versions

- **Advanced AI Models:** Use more powerful AI models to improve fake document detection accuracy.
- **More Document Types:** Support additional documents such as voter ID, certificates, visas, and international IDs.
- **Real-Time Verification:** Enable instant verification through secure government or organizational databases.
- **Blockchain Integration:** Store verification records securely using blockchain for better transparency and traceability.
- **Mobile Application:** Develop a mobile app for easy document submission and verification.
- **Liveness Detection:** Add advanced facial liveness detection to prevent photo and video-based identity fraud.
- **Multilingual Support:** Support multiple Indian languages for wider accessibility.
- **Continuous Learning:** Update AI models regularly to detect new and sophisticated fraud techniques.